

Technical Indicators & Strategy Backtesting with Alpaca: AAPL

Strategy Descriptions and Rules

Buy & Hold: Invests in the selected ticker throughout the full sample period.

Trend Following: Buys when MACD is above its signal line, ADX is greater than 25, and SMA20 is above SMA50. Sells when MACD falls below signal or SMA20 falls below SMA50.

Mean Reversion: Buys when RSI is below 30 and price is below the lower Bollinger Band. Sells when RSI is above 70 or price is above the upper Bollinger Band.

Custom Strategy: Combines trend, momentum, and volume. Buys when price is above SMA50, MACD is above signal, RSI is between 45 and 70, CMF is positive, and OBV is above OBV SMA20. Sells on trend weakness or extreme RSI conditions.

Performance Comparison

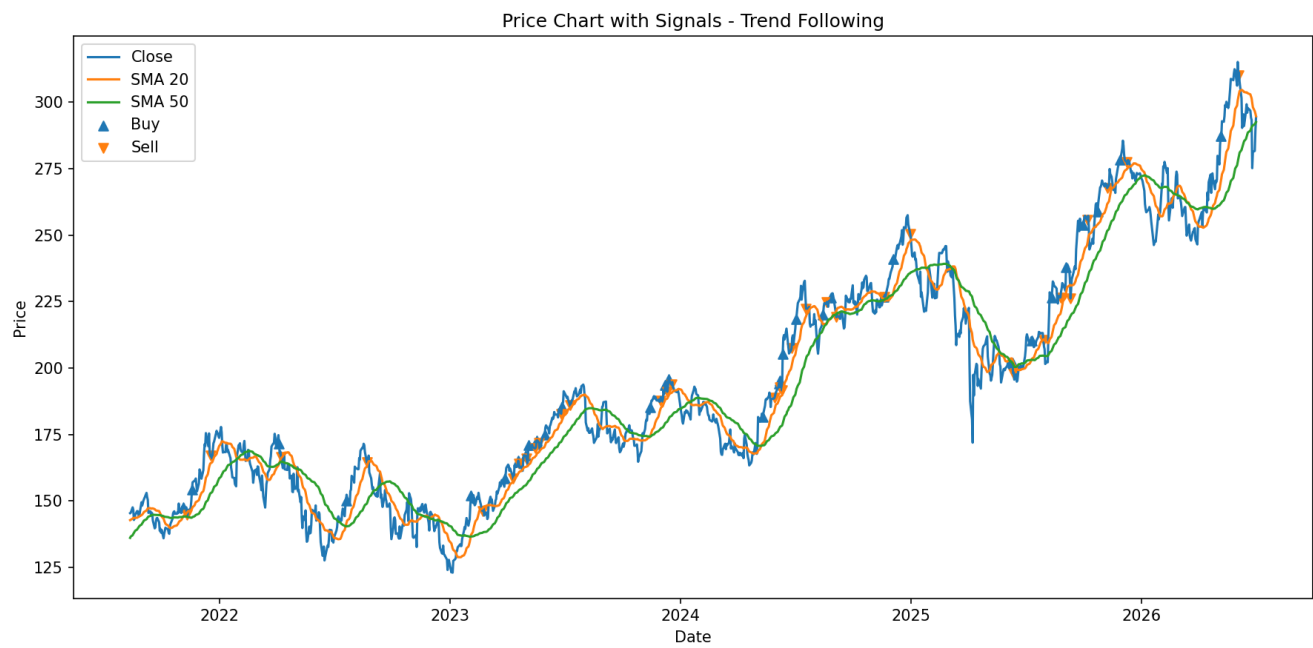
Strategy	Total Return	CAGR	Volatility	Sharpe Ratio	Sortino Ratio	Max Drawdown	Win Rate	Trades
Buy & Hold	102.57%	15.62%	27.75%	0.66	0.96	-33.25%	N/A	1
Trend Following	23.73%	4.47%	9.12%	0.53	0.39	-9.78%	50.00%	32
Mean Reversion	92.10%	14.36%	18.00%	0.83	0.62	-24.07%	85.71%	15
Custom Strategy	14.06%	2.74%	9.73%	0.33	0.23	-23.56%	43.75%	48

Discussion of Results

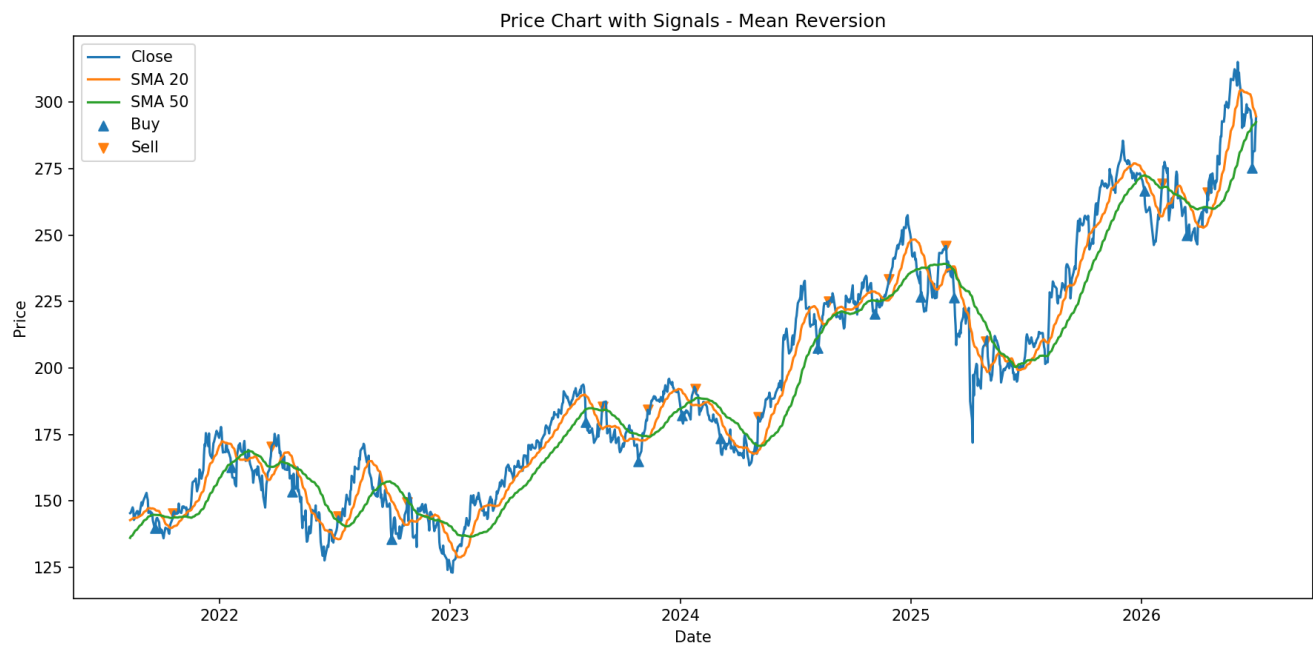
Based on risk-adjusted performance, the best strategy in this backtest is Mean Reversion with a Sharpe Ratio of 0.83 and a Sortino Ratio of 0.62. For the AAPL backtest, the Mean Reversion strategy performs best on a risk-adjusted basis because it has the highest Sharpe Ratio and Sortino Ratio among the tested strategies. This suggests that buying during oversold conditions and exiting during overbought conditions worked better for this ticker and sample period than the trend-following and custom strategies. The final conclusion depends on the selected ticker and sample period. Because the engine uses previous-day positions for next-day returns, the backtest avoids look-ahead bias. However, this simplified project does not include transaction costs, slippage, taxes, or survivorship-bias controls.

Charts

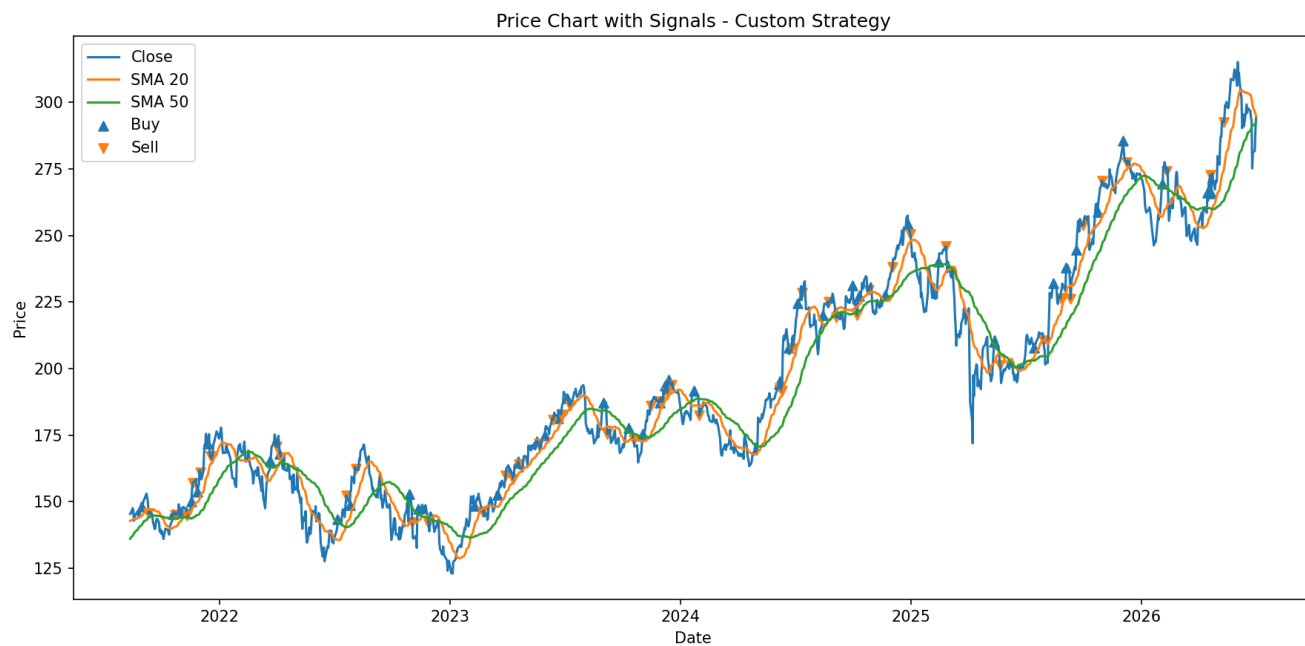
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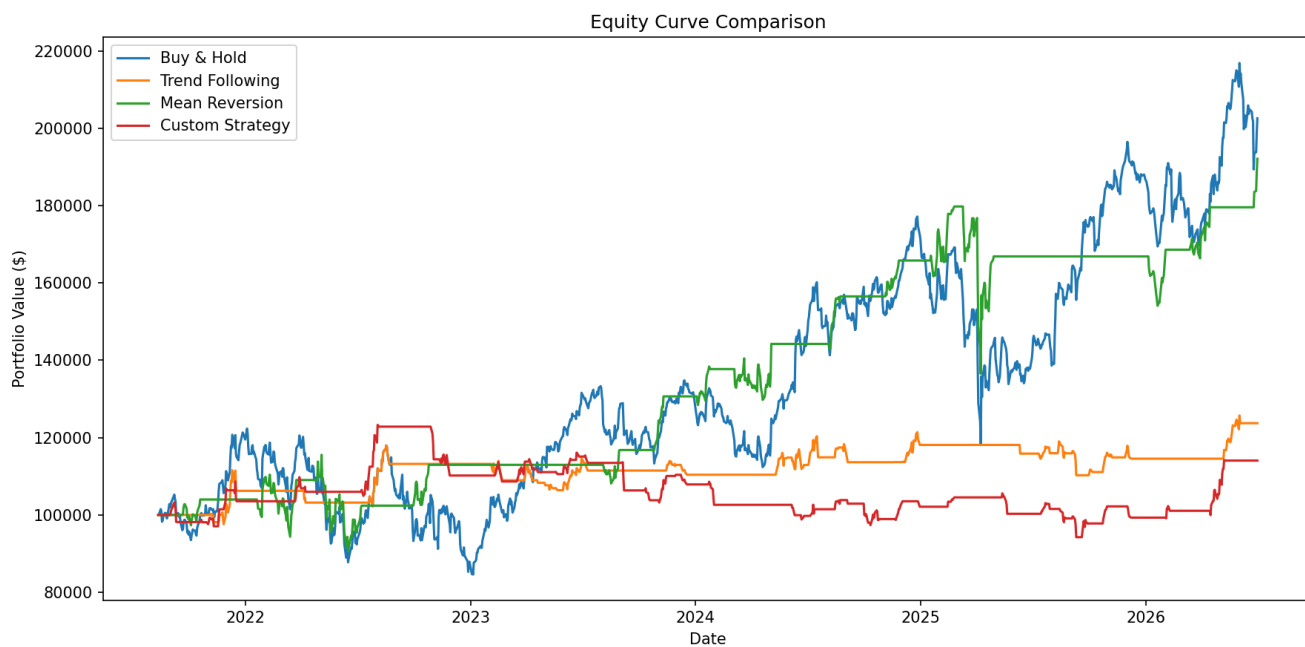
price_signals_mean_reversion.png



price_signals_custom_strategy.png



equity_curves.png



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